SIMATS ENGINEERING
ASSESSMENT TOOL 1
Experiments

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN

COURSE CODE: CSA1153

COURSE OUTCOME COVERED

CO3: Analyze system requirements using object-oriented techniques to identify classes, attributes, and relationships and develop use case models. (BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 4 **Total Marks:50**

Annexure A

Experiments

(Reg. No. 192411219)

a) Develop a system using UML for implementing a Library Management System. The system should allow the librarian to add and manage books through the central database. The student can search for books and request to borrow them. The system verifies availability and records the issue and return dates. Overdue fines are calculated and notified to the student automatically.

b) Develop a system using UML for implementing a Hospital Appointment System. The patient first registers through the system and books an appointment with the desired doctor. The central system verifies doctor availability and confirms the appointment. The receptionist updates the schedule and the doctor accesses the appointment list prior to the consultation.

(a) Library Management System – Use Case Diagram

Aim

To develop a UML Use Case Diagram for a Library Management System that manages books, borrowing, returns, availability verification, and overdue fine notifications.

Actors

1. Librarian
2. Student
3. Notification System (External Actor)

Use Cases

Librarian

- Login
- Add Book
- Update Book Details
- Remove Book
- Manage Database
- Issue Book
- Accept Return

Student

- Login
- Search Book
- Request Book
- Borrow Book
- Return Book
- View Fine Details

Notification System

- Send Fine Notification
- Send Due Date Reminder

Relationships

<<include>>

- Borrow Book <<include>> Verify Availability
- Return Book <<include>> Calculate Fine
- Add Book <<include>> Update Database
- Request Book <<include>> Search Book

<<extend>>

- Send Fine Notification <<extend>> Calculate Fine
- Send Due Date Reminder <<extend>> Borrow Book

Generalization

- Librarian and Student inherit from User

Use Case Diagram (Text Representation)

```
    LIBRARY MANAGEMENT SYSTEM

           User
            ^
    -----------------
    |               |
 Student        Librarian
```

Student

```
----------------------------------------
Login
Search Book
Request Book
Borrow Book
Return Book
View Fine Details
----------------------------------------
```

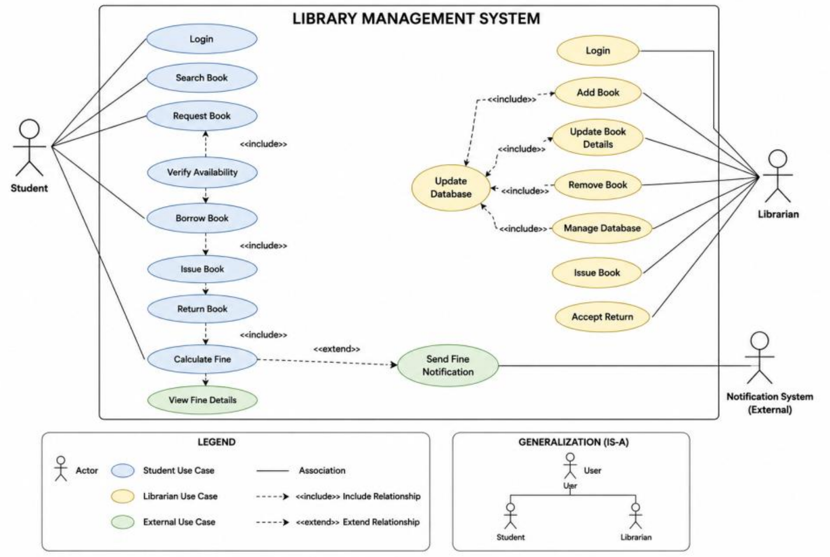
Librarian

```
----------------------------------------
Login
Add Book
Update Book
Remove Book
Manage Database
Issue Book
Accept Return
----------------------------------------
```

Notification System

```
----------------------------------------
Send Fine Notification
Send Due Date Reminder
----------------------------------------
```

Borrow Book <<include>> Verify Availability
Return Book <<include>> Calculate Fine
Send Fine Notification <<extend>> Calculate Fine



Conclusion

The Library Management System Use Case Diagram was successfully designed by identifying the actors Librarian, Student, and Notification System. The diagram models book management, borrowing, returning, fine calculation, and notification processes, ensuring efficient library operations and better user interaction.

(b) Hospital Appointment System – Use Case Diagram

Aim

To develop a UML Use Case Diagram for a Hospital Appointment System that enables patient registration, appointment booking, doctor availability verification, and schedule management.

Actors

1. Patient
2. Receptionist
3. Doctor

Use Cases

Patient

- Register
- Login
- Search Doctor
- Book Appointment
- View Appointment Status

Receptionist

- Update Schedule
- Manage Appointments
- Confirm Appointment

Doctor

- View Appointment List
- Manage Consultation Schedule

Relationships

<<include>>

- Book Appointment <<include>> Verify Doctor Availability
- Confirm Appointment <<include>> Update Schedule
- Register <<include>> Create Patient Record

<<extend>>

- View Appointment Status <<extend>> Book Appointment
- Manage Consultation Schedule <<extend>> View Appointment List

Generalization

- Patient, Receptionist, and Doctor inherit from User

Use Case Diagram (Text Representation)

HOSPITAL APPOINTMENT SYSTEM

```
               User
                ^
     ----------------------------------------
     |            |              |
   Patient     Receptionist       Doctor
```

Patient

Register
Login
Search Doctor
Book Appointment
View Appointment Status

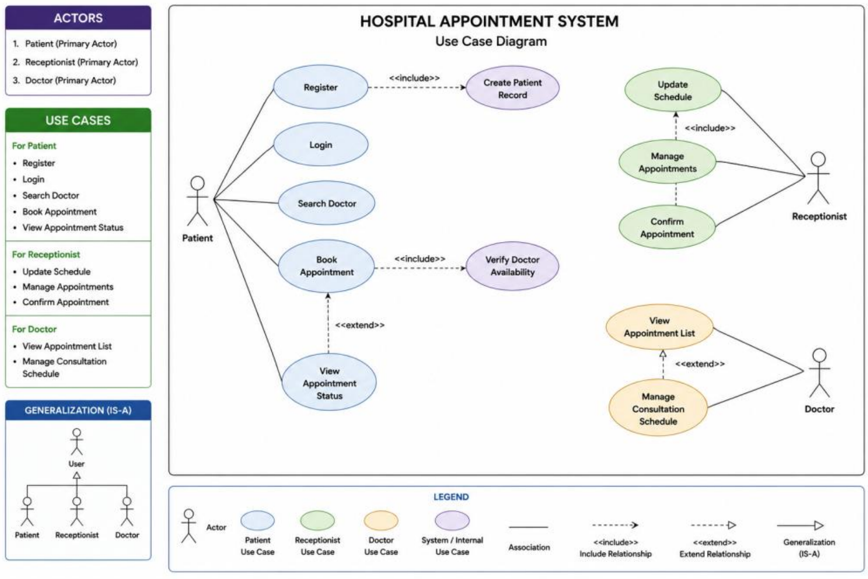
Receptionist

Update Schedule
Manage Appointments
Confirm Appointment

Doctor

View Appointment List
Manage Consultation Schedule

Book Appointment <<include>> Verify Doctor Availability
Confirm Appointment <<include>> Update Schedule
View Appointment Status <<extend>> Book Appointment



Conclusion

The Hospital Appointment System Use Case Diagram was successfully developed by identifying the actors Patient, Receptionist, and Doctor along with their interactions. The diagram provides a clear representation of appointment scheduling, availability verification, confirmation, and consultation management, ensuring smooth hospital workflow and effective patient service.

Code of Conduct:

I <u>Y. Abhay-192411219</u> (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
Y. Abhay

RUBRICS FOR CALCULATION:

Experiments

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation